
HARNESS VM DEPLOYMENT – SERVICE-TO-HOST FILTERING

Proposal & Implementation Guide

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SUMMARY

Currently, when deploying multiple services to VMs in Harness, the platform creates a cartesian product – it tries every combination of service and host. This causes incorrect deployments where services are pushed to VMs they are not meant to run on.

This document proposes a solution using Harness host attributes and dynamic host filtering so that each service deploys only to the VMs it belongs to, with an optional runtime toggle to bypass the filter when needed.

PROBLEM STATEMENT

Harness deploys every service to every VM when multiple services and multiple VMs are selected in a single pipeline stage.

Example of the current incorrect behaviour:

- Service A, Service B selected
- VM1, VM2, VM3 selected in infrastructure definition
- Harness deploys: A->VM1, A->VM2, A->VM3, B->VM1, B->VM2, B->VM3 (6 deployments)
- Expected: A->VM1, A->VM2, B->VM3 (only 3 deployments)

PROPOSED SOLUTION

Use Harness built-in Host Attributes on each VM to tag which services run on it, then apply a Host Filter expression in the Infrastructure Definition that automatically matches the deploying service to its correct VMs.

An optional pipeline-level toggle allows the filter to be bypassed at runtime when a full deployment to all VMs is needed.

HOW IT WORKS:

Component	What It Does
Host Attributes	Tags each VM with which services run on it
service.name variable	Built-in Harness variable – auto-resolves to service name
Host Filter Expression	Matches service name to host attributes at runtime
enableServiceFilter	Pipeline-level runtime toggle (dropdown) to turn filter on/off

IMPLEMENTATION STEPS

STEP 1 – TAG VMs IN INFRASTRUCTURE DEFINITION

Add host attributes to each VM indicating which services run on it.
Only add a tag when the value is true – missing attributes default to false.

VM	Attribute Key	Value	Meaning
vm1.prod.internal	auth	true	auth service runs here
vm1.prod.internal	payments	true	payments service runs here
vm2.prod.internal	payments	true	payments service runs here
vm3.prod.internal	notifications	true	notifications runs here

Steps in the UI:

1. Go to Deployments -> Environments
2. Select your environment (e.g. prod)
3. Click the Infrastructure Definitions tab
4. Open your infra def and click edit/pencil icon on each host
5. In Host Attributes, add key-value pairs for each service on that VM
6. Save the infra def

STEP 2 – ADD THE RUNTIME TOGGLE VARIABLE

Add a pipeline-level variable with allowed values so it renders as a dropdown at runtime.

Steps in the UI:

1. Go to Pipelines and open your pipeline
2. Click on the Pipeline itself (not a stage) in the pipeline studio

3. In the right panel, click Variables
4. Click Add Variable and configure:
 - Name: enableServiceFilter
 - Type: String
 - Value: click the thumbtack icon -> select Runtime Input
5. Click the gear/settings icon next to the variable
6. Select Allowed Values and add: true and false
7. Apply changes and save the pipeline

STEP 3 – SET THE HOST FILTER EXPRESSION

Paste the following expression into the Host Filter field of the Infrastructure Definition:

```
<+<+pipeline.variables.enableServiceFilter> == "true"
  ? <+infra.hostObject.hostAttributes[<+service.name>]> ? : "false"
  : "true"> == "true"
```

How this expression works:

- If enableServiceFilter = true -> checks host attribute matching service name
- If that attribute exists and is "true" -> VM is included
- If that attribute is missing or not "true" -> VM is excluded (defaults to false)
- If enableServiceFilter = false -> all VMs included, filter is bypassed

Steps in the UI:

1. Go back to Environments -> Infrastructure Definitions
2. Open your infra def
3. Find the Host Filter field (under Advanced section)
4. Paste the expression above
5. Save

STEP 4 – RUNNING THE PIPELINE

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1. Click Run Pipeline
 2. A dropdown will appear for enableServiceFilter with two options:
 - true -> filter is ON, each service deploys only to its tagged VMs
 - false -> filter is OFF, all services deploy to all VMs
 3. Select the option and click Run

KEY POINTS & GOTCHAS

- The host attribute key must match the Harness service name exactly (case-sensitive)
- Do not add false tags – simply omit the attribute on VMs where service does not run
- The enableServiceFilter toggle is pipeline-level, applies to all stages at once
- The Host Filter expression must be saved on the Infra Def, not the pipeline
- Service names with hyphens (e.g. my-auth-service) are supported – the attribute key just needs to match exactly

QUICK REFERENCE

What	Where in UI	Key
Detail		
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Host attributes (tags)	Environments -> Infra Def -> Host edit	Only add tags that are true
enableServiceFilter	Pipeline -> Variables	Runtime Input, Allowed Values: true/false
Host filter expression	Environments -> Infra Def -> Advanced	Paste the full expression
service.name	Built-in Harness variable	No setup needed

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Questions or feedback? Add comments to this document or raise in team channel.

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